

MDF++: Visuo-Tactile World Action Model with End-to-end Representation Learning

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Abstract: Recently, world-action models (WAMs) have emerged as a powerful framework for robot manipulation. By jointly modeling action generation and future observation prediction, WAMs can act and anticipate the outcomes of the actions. However, existing WAMs typically rely on frozen pretrained encoders trained with reconstruction, which may not produce representations well suited for robot control. In this paper, we tackle the problem of learning physically-grounded, control-relevant representations for WAMs. We propose MDF++, a visuo-tactile world-action model that learns latent representations of multisensory observations end-to-end with a diffusion backbone. While naive joint training can lead to feature collapse, MDF++ introduces a multi-task learning strategy that leverages cross-modal correlations to learn task-relevant representations. We evaluate MDF++ on a suite of contact-rich manipulation tasks, including real-world bolt screwing and nut mating, and show that it outperforms WAMs built on frozen pretrained representations by up to 50%. Additional videos are available on our website: jointmdf.github.io.

Keywords: Multimodal World Model, Contact-rich Manipulation

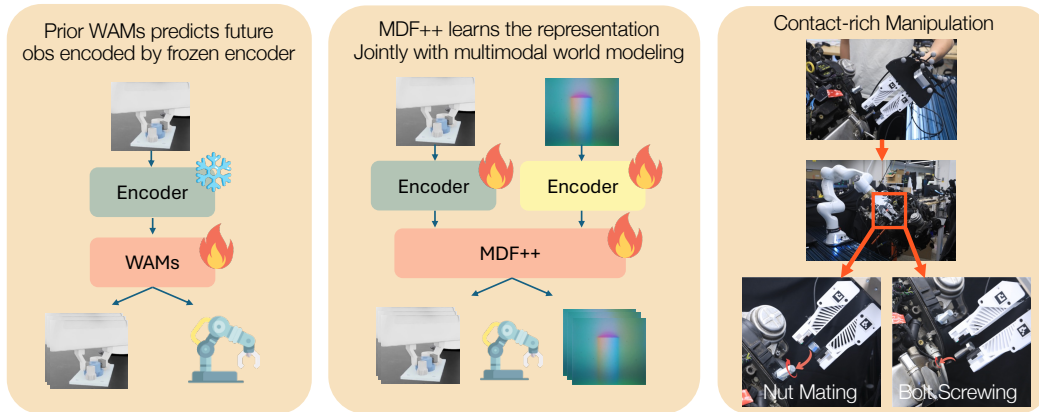


Figure 1: Existing WAMs typically predict future observations encoded by a frozen pretrained encoder optimized for visual generation or semantic understanding, which may be suboptimal for robot manipulation. We propose MDF++, a visuo-tactile world-action model that learns task-aware representations end-to-end with cross-modality prediction.

1 Introduction

When tightening a screw or mating a nut, vision alone is often insufficient: successful execution depends on subtle contact cues, force feedback, and proprioceptive information. In order to perform such tasks, it would also be useful to anticipate how the robot’s actions affect the world, such as how contact will form, how an object will move, and how force will propagate. This motivates robot

models that jointly reason about actions and their consequences. World-action models (WAMs) [1, 2, 3, 4, 5, 6, 7, 8, 9, 10] address this by jointly modeling future observations and actions within a single generative model. Recent work [2, 9, 10] shows that such auxiliary dynamics learning can improve model capability and generalization in robot control.

However, existing WAMs still have two key limitations for robot manipulation. First, most prior WAMs are built primarily around visual observations, leaving tactile and force information largely unexplored. Second, these models typically rely on frozen pretrained encoders, often trained with pixel reconstruction or self-supervised visual objectives. While such encoders provide useful features for image generation or semantic understanding, their representations are not necessarily aligned with robot control. This issue is even more pronounced for tactile, where broadly-applicable pretrained encoders are less established. A natural solution is to finetune the encoders end-to-end with the WAM objective, but we find that naive finetuning leads to feature collapse: the encoder and diffusion backbone co-adapt to produce latents that are easy to denoise but less useful for control.

To address the above limitations, we present MDF++, a visuo-tactile world-action model that learns task-relevant representations end-to-end with a latent diffusion backbone. We build on Multimodal Diffusion Forcing (MDF) [10], but instead of modeling dynamics in a frozen latent space, MDF++ finetunes observation encoders jointly with the world-action model under a mixture of task-oriented objectives, including policy learning, joint action–observation prediction, forward dynamics modeling, and inverse dynamics modeling. To avoid feature collapse during joint training, MDF++ introduces a cross-modal training strategy with selective gradient back-propagation, which encourages the latent space to capture the interplay between different modalities, yielding representations that are better aligned with robot control.

Our key contributions are three-fold. First, we propose MDF++, a diffusion-based visuo-tactile WAM that learns task-relevant multimodal representations end-to-end using cross-modal prediction. Second, we introduce a selective gradient-flow strategy that enables stable joint training of observation encoders and the world-action model while avoiding feature collapse. Third, we evaluate MDF++ on seven contact-rich manipulation tasks, including two real-world threading tasks requiring millimeter-level precision. In simulation, MDF++ outperforms frozen-encoder baselines by 28% and naive joint training by 22%. In real, it outperforms video-based WAMs [9] by 29%.

2 Related Work

2.1 World modeling for Robotics

Recently, world models have shown great promise for robotics. One line of work studies action-conditioned world models (WAs) to predict the future outcomes of actions, enabling data generation [11, 12], planning [13, 14, 15, 16], and policy evaluation [17, 12, 18]. Another line of work, world action models (WAMs), jointly predicts future observations and actions [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 19]. However, most prior work is limited to vision, whereas MDF++ captures the dynamics of multimodal sequences including tactile, force and robot proprioception.

Beyond the choice of modalities, a key design question is how to obtain the representation space in which world-action dynamics are learned. Most existing WA and WAMs rely on frozen pretrained encoders, typically trained with pixel reconstruction [20] or self-supervised learning objectives [21, 22]. For example, UWM [9] and UVA [2] use pretrained image VAEs [23, 20], Dino-WM [14] predicts future states in the DINO feature space [21], and MDF [10] builds on a diffusion-based point-cloud VAE [24]. While these encoders provide useful features for visual generation or semantic understanding, they are not necessarily aligned with robot control. As a result, the world model may learn dynamics in a representation space that is suboptimal for manipulation. In contrast, MDF++ starts from a pretrained encoder and finetunes it end-to-end with the latent world-action model backbone. Through cross-modal supervision from tactile, force, and actions, MDF++ learns a representation space that is better suited for robot manipulation. FLARE [25] finetunes the encoder using an EMA target, but it is not designed for unified WAMs with multiple query distributions and also does not support multimodal reasoning.

72 2.2 Multimodal Representation learning for Manipulation

73 The integration of multi-sensory information, such as audio [26, 27, 28], force [29, 30, 31], is critical
 74 for robot control. Particularly, there is growing interest in combining vision and tactile for adaptive
 75 and precise manipulation. Much of the prior works learn a visuo-tactile policy using either imitation
 76 learning [32, 33, 34, 35, 36] or reinforcement learning [37, 38, 39, 40, 41, 42, 43], where representa-
 77 tions are primarily supervised by action prediction or task reward. Other works study self-supervised
 78 tactile [44, 45, 46, 35] or visuo-tactile representation learning [47, 36]. In contrast, our work studies
 79 representation learning for multimodal world-action models. While these models provide richer su-
 80 pervision through action and cross-modal prediction, naively training encoders with diffusion losses
 81 can cause feature collapse. We address this with selective gradient flow and show that the proposed
 82 method outperforms policy-only training and pretrained self-supervised encoders [45].

83 3 Preliminaries

84 3.1 Latent Diffusion Models

85 Diffusion models learn a data distribution by gradually corrupting clean samples with Gaussian
 86 noise and training a neural network to reverse this process [48]. Given a clean sample $\mathbf{x}_0$, the
 87 forward process produces a noisy sample $\mathbf{x}_k = \sqrt{\bar{\alpha}_k}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_k}\epsilon$ at diffusion step k , where
 88 $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\bar{\alpha}_k$ is determined by a noise schedule. The reverse model is trained to denoise
 89 $\mathbf{x}_k$ by predicting the injected noise, yielding the standard objective $\mathbb{E}_{k, \mathbf{x}_0, \epsilon} [\|\epsilon - \epsilon_\theta(\mathbf{x}_k, k)\|^2]$. At
 90 inference time, samples are generated by iterative denoising from Gaussian noise. Latent diffusion
 91 models perform this denoising process in a learned latent space rather than directly in the input
 92 space [23]. A sample $\mathbf{x}_0$ is first encoded into a latent representation $\mathbf{z}_0 = \mathcal{E}(\mathbf{x}_0)$, where the encoder
 93 $\mathcal{E}$ is typically pretrained and kept frozen during diffusion training.

94 3.2 Unified World-Action Model

95 A unified world-action model (WAM) is a single generative model over observations and actions
 96 that can be queried under different conditioning patterns. Let $\mathbf{o}_{i:j}$ and $\mathbf{a}_{i:j}$ denote observation
 97 and action sequences from timestep i to j , respectively. Given an observation history $\mathbf{o}_{t-h:t}$ and
 98 a prediction horizon H , the same model can be used as a policy, a forward dynamics model,
 99 an inverse dynamics model, or a joint action-observation predictor by choosing which variables
 100 are conditioned on and which variables are generated: **policy** $p(\mathbf{a}_{t+1:t+H} \mid \mathbf{o}_{t-h:t})$, **world-**
 101 **action model** $p(\mathbf{o}_{t+1:t+H}, \mathbf{a}_{t+1:t+H} \mid \mathbf{o}_{t-h:t})$, **world model / forward dynamics** $p(\mathbf{o}_{t+1:t+H} \mid$
 102 $\mathbf{o}_{t-h:t}, \mathbf{a}_{t+1:t+H})$, and **inverse dynamics model** $p(\mathbf{a}_{t+1:t+H} \mid \mathbf{o}_{t-h:t+H})$.

103 3.3 Problem Formulation

104 We consider diffusion-based world-action modeling for contact-rich robot manipulation from an
 105 offline dataset of expert demonstrations. At each timestep t , the robot receives a multimodal obser-
 106 vation $\mathbf{o}_t = \{\mathbf{o}_t^m\}_{m \in \mathcal{O}}$, where $\mathcal{O}$ denotes the set of observation modalities, such as vision, tactile
 107 sensing, force/torque sensing, and proprioception. The robot action is denoted by $\mathbf{a}_t$.

108 High-dimensional observations such as image and tactile are mapped into a latent representation by
 109 a modality-specific encoder ϕ_m : $\mathbf{z}_t^m = \phi_m(\mathbf{o}_t^m)$, where $m \in \mathcal{O}$. Together with the action stream,
 110 these latent observation tokens define a multimodal trajectory

$$\mathbf{x}_{1:T} = \{(\{\mathbf{z}_t^m\}_{m \in \mathcal{O}}, \mathbf{a}_t)\}_{t=1}^T.$$

111 We consider the problem of learning a unified world-action model from multimodal demonstration
 112 trajectories for contact-rich manipulation. Given observations from multiple sensory modalities, our
 113 goal is to train observation encoders jointly with a diffusion-based world-action model to capture
 114 task-relevant feature. At test time, we evaluate the model primarily through its ability to generate
 115 actions for robot manipulation tasks.

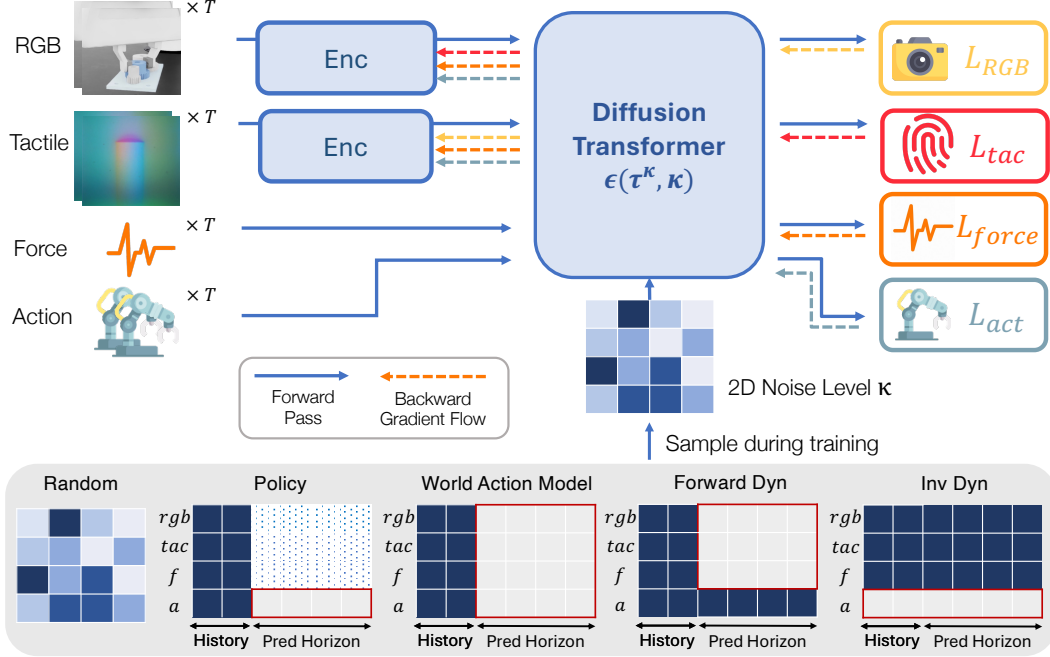


Figure 2: MDF++ training overview. MDF++ jointly trains observation encoders with a latent diffusion world-action model. During training, it samples structured 2D noise patterns for control-relevant tasks, including policy learning, world-action modeling, forward/inverse dynamics, and random masking, to capture temporal and cross-modal dependencies. To avoid collapsed representations that are easy to denoise but weak for control, MDF++ blocks each modality’s denoising loss from updating its own encoder. Instead, each encoder is trained through cross-modal prediction losses, encouraging task-relevant cues for predicting other sensory modalities and actions.

4 Joint Representation Learning for Unified World-Action Model

4.1 Multi-task Training with Multimodal Diffusion Forcing

To train a single backbone that supports the diverse conditional queries of a unified WAM (Sec. 3.2), we adopt *Multimodal Diffusion Forcing* (MDF) [10] as our model architecture. Standard diffusion models apply a single scalar noise level k to the entire sample [48], imposing a fixed separation between conditioning inputs and prediction targets. This is restrictive for unified world-action modeling, where one model must support multiple prediction patterns.

MDF addresses this by treating noise as a continuous form of masking [49, 10]: zero noise corresponds to an observed token, intermediate noise to a partially corrupted token, and maximum noise to a fully masked token. By assigning noise levels independently across time and modality, MDF can express different conditional distributions over the same trajectory. Concretely, consider a trajectory of latent multimodal tokens and actions over a horizon T , $\tau = \{(z_t, a_t)\}_{t=1}^T$, where each timestep contains M modality streams (i.e., vision, tactile, force, proprioception, and action). MDF defines a 2D *Time-Modality Noise Level Matrix* $\mathbf{K} \in \{0, \dots, K\}^{T \times M}$, where each entry $k_{t,m}$ specifies the diffusion step applied to modality m at time t . The forward process corrupts each token according to its own noise level, $\tau_{t,m}^{k_{t,m}} = \sqrt{\bar{\alpha}_{k_{t,m}}} \tau_{t,m}^0 + \sqrt{1 - \bar{\alpha}_{k_{t,m}}} \epsilon_{t,m}$, $\epsilon_{t,m} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. A diffusion transformer ϵ_θ conditioned on both the corrupted trajectory $\tau^{\mathbf{K}}$ and the matrix $\mathbf{K}$ is then trained to predict the per-token noise with the standard DDPM objective [48],

$$\mathcal{L}_{\text{MDF}}(\theta) = \mathbb{E}_{\tau^0, \mathbf{K}, \epsilon} \left[\sum_{t=1}^T \sum_{m=1}^M \|\epsilon_{t,m} - \epsilon_\theta(\tau^{\mathbf{K}}, \mathbf{K})_{t,m}\|^2 \right]. \quad (1)$$

While the original MDF training scheme is highly flexible, it samples the noise-level matrix $\mathbf{K}$ from a purely random distribution (bottom left of Fig. 2). This covers a large family of prediction tasks,

since each masking pattern defines a different conditional distribution. However, many of these conditionals are only weakly related to the inference problems needed for robot manipulation. To bias training toward control-relevant queries, we train the model on five structured conditional distributions with equal sampling probability: **policy learning**, **world-action modeling**, **forward dynamics**, **inverse dynamics**, and **random masking**. As shown in Fig. 2, conditioning tokens are assigned noise level zero, while prediction tokens are assigned noise levels sampled uniformly from $[0, K]$. These structured patterns focus model capacity on conditional distributions directly useful for robot control. Although our final downstream objective is action generation for visuomotor control, prior work shows that auxiliary prediction tasks, such as forward and inverse dynamics modeling, can encourage representations to capture task-relevant features and temporal structure in demonstration trajectories [50, 51, 52].

However, **existing WAMs typically cannot fully leverage this multi-task supervision for representation learning**, because they rely on frozen encoders pretrained with reconstruction objectives [53, 2, 9, 10]. For example, MDF [10] uses a frozen point-cloud encoder pretrained with a reconstruction objective. Reconstruction encourages the encoder to preserve information needed to reproduce the input, including task-irrelevant details that may not be useful for downstream control. While this design provides stable geometric features, it decouples representation learning from the control and prediction objectives used by the world-action model. In the next section, we describe how MDF++ leverages this multi-task supervision to train observation encoders end-to-end with the diffusion backbone, yielding action-aware representations for robotic manipulation.

4.2 Cross-modal Training with Selective Gradient Flow

Given a diffusion-based WAM, the most direct way to fine-tune the observation encoders is to back-propagate the diffusion loss through the encoder. However, this naive end-to-end training scheme usually leads to collapsed feature space. Since the same latent tokens serve as both conditioning inputs and prediction targets, gradients from a modality’s own denoising loss can encourage the encoder and diffusion backbone to co-adapt toward a latent space that is easy to denoise [54], but weak for control. In our experiments, this direct fine-tuning strategy often degrades task performance.

MDF++ avoids feature collapse with a simple selective gradient-flow rule inspired by REPA-E [54]: the encoder for each observation modality is updated only by diffusion losses from other streams, not by the loss of its own stream. Let $\mathcal{O}$ denote the set of observation modalities and $\mathcal{M} = \mathcal{O} \cup \{a\}$ denote all predicted streams, including actions. Let $\mathcal{L}_s$ be the MDF denoising loss for stream $s \in \mathcal{M}$. For each observation encoder ϕ_o , where $o \in \mathcal{O}$, MDF++ applies the masked gradient

$$\nabla_{\phi_o}^{\text{MDF++}} \mathcal{L} = \sum_{s \in \mathcal{M}} \mathbf{1}[s \neq o] \nabla_{\phi_o} \mathcal{L}_s. \quad (2)$$

Equivalently, when computing the denoising loss for modality m , gradients are stopped at the output of its own encoder ϕ_m , while gradients from cross-modal prediction losses are allowed to update it.

This rule trains each encoder to produce representations that are predictive of future signals in other modalities. For example, the visual encoder is not optimized to reconstruct visual appearance itself; instead, it is optimized through losses for predicting other sensory streams and actions. Since predicting tactile feedback, interaction forces, and successful actions requires information about the object, the end-effector, and their potential contact regions, this cross-modal supervision naturally encourages the visual representation to capture manipulation-relevant cues. Because no encoder is trained to denoise its own input, the learned representations are shaped by cross-modal predictive pressure rather than modality-specific reconstruction shortcuts.

Model Architecture. We instantiate τ as a sequence of modality-specific tokens. RGB and tactile images are encoded as CLS tokens from pretrained ViTs [55], while low-dimensional streams such as force, proprioception, and actions are projected with linear layers. At each timestep, all modality tokens are concatenated with their corresponding noise-level embeddings and processed by a diffusion transformer. **Training Recipe.** We identify several design choices that are important for training stability and performance. First, for each sampled trajectory, we sample multiple train-

ing modes to encourage robust learning across different conditioning and prediction settings and improve training efficiency. Second, MDF++ injects different noise levels into latent tokens during diffusion training, we detach the encoder outputs for highly corrupted tokens; gradients from these samples are often weakly correlated with the original observations and can destabilize representation learning. Third, we apply a cosine learning-rate schedule to the encoder and cool it down before the diffusion backbone.

5 Experiments

We evaluate MDF++ on a suite of simulated and real-world visuo-tactile manipulation tasks. Our experiments are designed to answer three questions: (1) Does the proposed selective gradient-flow strategy stabilize joint representation learning? (2) Are the learned task-aware representations more effective than pretrained semantic or reconstruction-based encoders for contact-rich manipulation? (3) Does multi-task training provides a better representation for contact-rich robot manipulation?

5.1 Simulation Experiments

Tasks: We consider five manipulation tasks that requires multimodal reasoning. Three precision-focused tasks are selected from Isaac Lab FACTORY [56]: PegInsert, GearMesh, and NutThread. We adapt the simulation environments from Forge [29] by adding a GelSight Mini tactile sensor. For each task, we collect 1,000 demonstrations using an RL expert policy. We additionally include two tactile-exploration tasks from MANIFEEL [57]. In ObjectSearch, the robot must locate, grasp, and lift a cube-shaped object randomly placed inside a closed container. In BallSorting, the robot must sort two visually similar balls using tactile sensing.

Baselines and ablations We compare MDF++ against several variants and representative baselines. **MDF++ (naive)** jointly trains the encoders and diffusion model by backpropagating denoising losses from all modalities through all encoders. This baseline tests whether direct end-to-end diffusion training is sufficient for representation learning. **MDF++ (SIGReg)** adds SIGReg [58, 15] as a latent-space regularizer to mitigate representation collapse. SIGReg optimizes a normality test over random one-dimensional projections, encouraging the latent embeddings to match an isotropic Gaussian distribution. **MDF++ (frozen-ViT)** and **MDF++ (frozen-Sparsh)** use frozen pretrained encoders for visual and tactile observations. The pretrained ViT [59] provides a semantic representation, while Sparsh [45] is pretrained specifically for vision-based tactile sensing. These baselines test whether fixed pretrained representations are sufficient for visuo-tactile control. **UWM** [9] is a unified world-action model that encodes future observations using a frozen SDXL VAE [20], which is optimized for high-quality pixel reconstruction. This baseline evaluates a reconstruction-oriented latent space for world-action modeling. Finally, **DP w/ tactile** is a diffusion policy baseline that conditions on both visual and tactile observations. This baseline compares MDF++ against a policy-only model without the proposed multimodal world-action training objective.

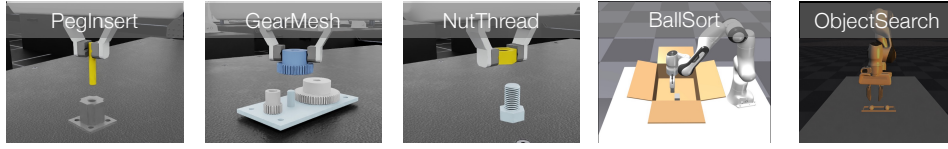
5.1.1 Results

The success rate of all methods can be found at Table. 1. The success rates are computed over 150 trials for each method.

Selective gradient flow stabilizes end-to-end representation learning. Table 1 shows that MDF++ achieves the best average success rate across the five simulated tasks, reaching 86.9%. In contrast, MDF++(naive), which directly backpropagates all denoising losses through all encoders, drops to 64.1%. This large gap supports our hypothesis that naive end-to-end diffusion training can produce shortcut latent representations that are easy to denoise but less useful for control. Adding SIGReg as a latent-space regularizer does not resolve this issue, achieving only 58.1% average success, suggesting that the regularization is insufficient for joint encoder learning of latent diffusion models.

Task-aware representations outperform frozen pretrained encoders. MDF++ substantially outperforms MDF++(frozen-ViT), MDF++(frozen-Sparsh) and UWM, which achieve average success

Table 1: Simulation tasks and success rates on five simulated visuo-tactile manipulation tasks.



Model	PegInsert $\uparrow$	GearMesh $\uparrow$	NutThread $\uparrow$	BallSort $\uparrow$	ObjectSearch $\uparrow$	Avg. $\uparrow$
MDF++	85.3	78.7	92.0	93.3	85.3	86.9
MDF++(naive)	56.7	50.0	90.7	67.3	56.0	64.1
MDF++(SIGReg [15])	50.7	32.0	84.7	48.0	75.3	58.1
MDF++(frozen-ViT)	86.7	66.0	89.3	36.7	14.7	58.7
MDF++(frozen-Sparsh)	90.0	64.7	81.3	40.0	18.7	58.9
MDF++(policy)	62.7	64.7	90.7	90.0	86.6	78.9
UWM [9] w/ tactile	88.6	72.0	89.3	53.3	68.0	74.2
DP [36] w/ tactile	88.7	81.3	91.3	28.0	54.0	68.7

rates of 58.7%, 58.9% and 74.2%, respectively. Although ViT and Sparsh provide strong semantic or tactile representations, their features are not optimized for the action, force, and contact dynamics required by these tasks. UWM performs better than the frozen-encoder ablations on average, but still lags behind MDF++. This suggests that predicting future observations at the pixel level can impose unnecessary reconstruction burden, forcing the model to allocate capacity to visual details that may be irrelevant for control. This limitation is especially evident on the tactile-exploration tasks: MDF++ achieves 93.3% on BallSort and 85.3% on ObjectSearch, while the frozen-encoder variants perform much worse. These results indicate that contact-rich manipulation benefits from representations shaped by task-specific multimodal prediction objectives.

World-action training outperforms policy-only learning across task types. Policy-only methods, including MDF++(policy) and DP w/ tactile, can learn strong reactive behaviors, especially on precision assembly tasks. For example, DP w/ tactile slightly outperforms MDF++ on PegInsert and GearMesh. However, policy-only training lacks explicit supervision for modeling how contacts and observations evolve under robot actions, which limits robustness on tasks requiring tactile exploration. MDF++ improves the average success rate over MDF++(policy) from 78.9% to 86.9% and over DP w/ tactile from 68.7% to 86.9%, with especially large gains on BallSort and ObjectSearch. These results suggest that MDF++’s dynamics-learning objectives provide richer supervision than action prediction alone, encouraging representations that capture action consequences and support both contact-rich assembly and tactile-driven exploration.

5.2 Physical Experiments

Hardware Platforms. We run our experiments on a Franka Research 3 robot with WSG-50 gripper. We use gelsight mini for tactile sensing and only feed the image of right finger to the policy. A Realsense 405 camera is mounted on the wrist to provide visual observation.

Tasks We evaluate our method on two challenging contact-rich manipulation tasks: Bolt Screwing and Nut Mating, on a real car engine. Both tasks require precise alignment under partial observability, where small pose errors can lead to failed insertion or jamming. In Bolt Screwing, the robot must align a bolt with a threaded hole and generate the appropriate downward force and rotational motion to initiate and maintain thread engagement. In Nut Mating, the robot must align a nut with a fixed bolt and guide it into the correct threading configuration. These tasks are particularly difficult because successful execution depends on millimeter-level accuracy; even slight misalignment can prevent the threads from engaging or cause cross-threading.

Data Collection and Processing Collecting high-quality demonstrations for Bolt Screwing and Nut Mating through direct teleoperation is challenging, as both tasks require precise alignment and axis-constrained rotation [60]. We therefore collect demonstrations using a customized version of UMI [61] equipped with GelSight sensors. In total, we collect 255 demonstrations for Nut Mating and 300 demonstrations for Bolt Screwing.

The original UMI formulation uses future end-effector states as actions to compensate for control delay. However, we find that this is insufficient for our tasks, since it does not account for controller tracking error. This issue is especially problematic for contact-rich manipulation, where millimeter-level accuracy is required, but blindly increasing controller gains may lead to excessive contact forces and potential hardware damage. To address this, we replay the collected trajectories in the robot frame and update the commanded actions using a residual correction proportional to the tracking error. Empirically, this reduces the average tracking error from approximately 4 mm to 1.2 mm, which is critical for successful execution. Please see appendix for details.

Evaluation protocol. At initialization, the nut or bolt is randomly placed between the robot fingers at different regions of the GelSight sensor, and each policy is evaluated under diverse initial end-effector poses. We evaluate three policies on two real-world threading tasks with 20 trials per policy-task pair. Each rollout is scored using a stage-based protocol: successful alignment receives 60 points, successful 180° rotation receives 75 points, and full credit is awarded only when the gripper releases and the nut or bolt remains fixed. The complete scoring rubric is provided in the appendix.

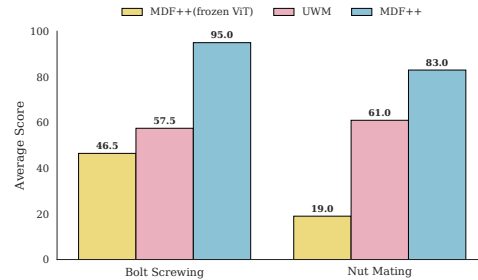


Figure 3: Real-world scores. MDF++ outperforms baselines on both threading tasks.

Result Analysis Across both Bolt Screwing and Nut Mating, MDF++ is robust to initial-configuration variation. The policy adapts to unseen initial end-effector poses and different initial bolt or nut placements on the GelSight sensor, and can also recover from small initial misalignments by making contact, refining alignment, and continuing rotation. In contrast, UWM is slower and often fails at phase transitions, stopping before reaching the object or rotating before thread engagement. We attribute this to UWM’s need to predict future visual observations at the pixel level, which becomes particularly challenging in the real-world engine environment due to its cluttered geometry, complex appearance, and precise contact and alignment requirements. MDF++(frozen-ViT) frequently stops in the air at the beginning of execution or after coarse alignment, occurring in 12 out of 20 Nut Mating trials. This suggests that frozen semantic visual features are insufficient for grounding high-precision manipulation actions in the complex engine environment.

6 Conclusion

We presented MDF++, a visuo-tactile world-action model that learns task-aware representations end-to-end with cross-modality prediction. By using multi-task world-action objectives together with selective gradient flow, MDF++ avoids pixel-level reconstruction and predicts future in a representation space that is helpful for manipulation. Experiments on simulated and real-world contact-rich manipulation tasks show that MDF++ outperforms frozen pretrained encoders, naive end-to-end finetuning, and policy-only training.

7 Limitations and future works

While MDF++ shows promising results for contact-rich manipulation, several limitations remain. First, MDF++ processes raw force signals at a relatively low frequency, which may miss high-frequency contact dynamics. Future work could learn force encoders over streaming high-frequency force measurements. Second, MDF++ requires manually tuning the loss weights for different modality. Automatically balancing these objectives would reduce task-specific tuning. Finally, although selective gradient flow helps prevent feature collapse, it also blocks gradients from each modality’s own denoising loss, potentially limiting the diffusibility of the learned latent space. Future work could explore alternative latent-training strategies, such as Unified Latents [62], to improve diffusibility while preserving control-relevant features.

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